

Temporal Lobe

Anatomy

Boundaries

The temporal lobe occupies the inferolateral aspect of each cerebral hemisphere. Its superior border is the **lateral (Sylvian) fissure**, which separates it from the frontal and parietal lobes. Posteriorly, the temporal lobe transitions into the occipital lobe along a less distinct boundary; on the lateral surface the **temporo-occipital line** (an arbitrary line from the preoccipital notch to the parieto-occipital sulcus) marks the convention, and on the medial surface the boundary follows the collateral sulcus and isthmus of the cingulate gyrus. The **anterior temporal pole** is the rostral tip.

Major gyri and sulci

The lateral surface is organized by the superior and inferior temporal sulci into three parallel gyri running anteroposteriorly:

- **Superior temporal gyrus (STG)** — above the superior temporal sulcus; contains auditory cortex and (in the dominant hemisphere) Wernicke's area
- **Middle temporal gyrus (MTG)** — between the superior and inferior temporal sulci; involved in language semantics, multimodal integration, and biological motion perception
- **Inferior temporal gyrus (ITG)** — below the inferior temporal sulcus; ventral visual stream and object recognition

The **superior temporal plane** (the upper surface of the STG, hidden within the Sylvian fissure) contains:

- **Heschl's gyrus (transverse temporal gyrus)** — primary auditory cortex
- **Planum temporale** — auditory association cortex, posterior to Heschl's; typically larger on the left and a critical substrate of Wernicke's area
- **Planum polare** — anterior to Heschl's

The **inferior (basal) surface** contains, from lateral to medial:

- **Inferior temporal gyrus** (wrapping from lateral surface)
- **Fusiform (occipitotemporal) gyrus** — ventral visual stream specializations including face and word recognition
- **Parahippocampal gyrus** — bounded medially by the hippocampal sulcus; contains

entorhinal and perirhinal cortices anteriorly

- **Uncus** — the medially curled anterior end of the parahippocampal gyrus, overlying the amygdala

Functional subdivisions

Region	Location	Primary role
Primary auditory cortex (A1)	Heschl's gyrus	Tonotopic processing of sound
Auditory association cortex	Superior temporal plane and STG surrounding Heschl's	Pitch, timbre, complex sound and music processing
Wernicke's area (dominant hemisphere)	Posterior STG, extending onto planum temporale	Auditory-verbal comprehension, phonological processing
Anterior temporal lobe (ATL)	Temporal pole and anterior MTG/ITG	Semantic memory hub, person and concept knowledge
Fusiform face area (FFA)	Mid-fusiform gyrus, typically right-lateralized	Face recognition, expert visual category processing
Visual word form area (VWFA)	Left mid-fusiform/occipitotemporal	Orthographic word recognition
Parahippocampal place area (PPA)	Parahippocampal gyrus	Scene and place recognition, spatial context
Hippocampus	Medial temporal lobe, within the temporal horn of the lateral ventricle	Episodic and relational memory encoding and retrieval
Entorhinal cortex	Anterior parahippocampal gyrus	Gateway to hippocampus; convergent input from neocortex
Perirhinal cortex	Lateral to entorhinal, along the rhinal sulcus	Object and item familiarity, recognition memory
Amygdala	Anterior medial temporal lobe, deep to uncus	Emotional processing, fear learning, salience
Superior temporal sulcus (STS)	Sulcus between STG and MTG	Biological motion, gaze and face dynamics, social perception

Vascular supply

- **Middle cerebral artery (MCA)** — lateral temporal cortex including STG (Wernicke's area), MTG, and lateral ITG; inferior division supplies the temporal lobe
- **Posterior cerebral artery (PCA)** — medial and inferior temporal lobe, including hippocampus, parahippocampal gyrus, fusiform gyrus, and uncus; also supplies temporal pole via anterior temporal branches
- **Anterior choroidal artery** — supplies the uncus, anterior hippocampus, and amygdala; small infarcts can produce memory and emotional changes

Key white matter tracts

- **Inferior longitudinal fasciculus (ILF)** — runs anteroposteriorly through the temporal lobe connecting occipital cortex to anterior temporal lobe; supports visual object recognition and visual-semantic integration
- **Uncinate fasciculus** — hooks from anterior temporal lobe to orbitofrontal and ventrolateral prefrontal cortex; semantic memory, social-emotional processing
- **Arcuate fasciculus / superior longitudinal fasciculus** — connects posterior temporal cortex (Wernicke's) with inferior frontal cortex (Broca's); damage produces conduction aphasia
- **Inferior fronto-occipital fasciculus (IFOF)** — long association tract from occipital and posterior temporal cortex to inferior frontal cortex; semantic processing
- **Fornix** — major output tract of the hippocampus to mammillary bodies, anterior thalamus, and septal nuclei (part of the Papez circuit); lesions produce amnesia
- **Cingulum** — runs above the corpus callosum and curves into the parahippocampal gyrus; connects medial temporal memory structures to cingulate and frontal cortex

Developmental note

The medial temporal lobe undergoes rapid early development, supporting the emergence of explicit memory across the second year of life. The hippocampus shows continued structural maturation into adolescence, and the protracted development of MTL-prefrontal connectivity is relevant to the maturation of strategic memory and source monitoring across childhood and adolescence.

Function

Auditory processing

- **Primary auditory cortex (A1)** is tonotopically organized; processes basic acoustic features

(frequency, intensity, temporal structure)

- **Auditory association cortex** supports recognition of complex sounds, music, environmental sounds, and prosody
- **Right STG** shows relative specialization for music, pitch, and prosody
- **Left STG** shows relative specialization for speech sound (phonological) processing

Language comprehension (dominant hemisphere)

- **Wernicke's area** supports auditory-verbal comprehension and phonological-to-lexical mapping
- **MTG and ITG** contribute to lexical-semantic retrieval and word meaning
- **Anterior temporal lobe** functions as a semantic memory hub, integrating multimodal conceptual knowledge

Memory

The medial temporal lobe (MTL) memory system supports declarative (explicit) memory:

- **Hippocampus** — encoding and retrieval of episodic memories; binding of multimodal features into coherent representations; spatial/relational memory
- **Entorhinal cortex** — major bidirectional gateway between hippocampus and neocortex
- **Perirhinal cortex** — item-level familiarity and recognition; visual object memory
- **Parahippocampal cortex** — scene and contextual memory

The MTL system is selectively involved in declarative memory; procedural, motor, and emotional learning depend on other systems (basal ganglia, cerebellum, amygdala).

Visual recognition (ventral stream)

The ventral occipitotemporal cortex constitutes the "what" pathway:

- **Object recognition** in lateral occipital and inferior temporal cortex
- **Face recognition** in fusiform face area
- **Place and scene recognition** in parahippocampal place area
- **Word recognition** in left occipitotemporal visual word form area

Social and emotional

- **Amygdala** — fear conditioning, emotional salience detection, modulation of memory for emotional events, processing of facial expressions of emotion
- **Superior temporal sulcus** — biological motion, gaze direction, facial movement, theory of mind

- **Anterior temporal lobe** — person-specific semantic knowledge, social concept representation

Lateralization

- **Left (dominant) temporal lobe** is preferentially engaged in language comprehension, verbal memory, and verbal semantic processing
 - **Right (non-dominant) temporal lobe** is preferentially engaged in visuospatial memory, face recognition, music processing, and prosody
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Clinical syndromes

Aphasia syndromes (dominant hemisphere)

Wernicke's aphasia

Fluent, well-articulated speech that is empty of content, with frequent **paraphasic errors** (phonemic and semantic) and **neologisms**. Auditory comprehension is impaired, repetition is impaired, and naming is impaired. Patients often lack awareness of their deficits. Lesion: posterior STG and adjacent supramarginal gyrus, MCA inferior division territory.

Transcortical sensory aphasia

Fluent with impaired comprehension but **preserved repetition**. Lesion: temporoparietal watershed, sparing perisylvian cortex. Patients may echo what is said without understanding it.

Conduction aphasia

Fluent speech with relatively preserved comprehension but **disproportionate impairment of repetition**, particularly for unfamiliar phrases. Phonemic paraphasias are prominent. Lesion: arcuate fasciculus or supramarginal gyrus.

Pure word deafness

Selective impairment of auditory-verbal comprehension with preserved reading, writing, and recognition of nonverbal sounds. Lesion: bilateral STG or unilateral dominant subcortical lesion disconnecting A1 from Wernicke's area.

Semantic variant primary progressive aphasia (svPPA / semantic dementia)

Progressive loss of semantic knowledge with anomia, impaired single-word comprehension, and surface dyslexia. Speech is fluent but empty. Lesion: bilateral anterior temporal atrophy.

surface dyslexia; spelling is intact but clumsy; Lesion: bilateral anterior temporal atrophy; typically left-predominant.

Memory disorders

Bilateral medial temporal lobe amnesia

The prototype is the case of **H.M.** (Henry Molaison), who developed dense **anterograde amnesia** with relatively preserved short-term memory, procedural learning, and remote memories following bilateral MTL resection. Hallmark features:

- Severe anterograde amnesia for declarative material
- Variable retrograde amnesia, typically with temporal gradient (Ribot's law)
- Preserved working memory and procedural learning
- Preserved semantic knowledge acquired before injury

Material-specific memory deficits

Unilateral MTL lesions produce lateralized memory impairments:

- **Left MTL** — verbal memory deficits (word lists, story recall)
- **Right MTL** — visuospatial memory deficits (figure recall, spatial layouts, face recognition)

Transient global amnesia (TGA)

Sudden-onset, time-limited (typically < 24 hours) anterograde amnesia with repetitive questioning, preserved personal identity and remote memory. Hippocampal/MTL involvement implicated; benign course.

Diencephalic amnesia (e.g., Korsakoff syndrome)

Although not temporal in origin, mammillary body and anterior thalamic damage produces amnesia clinically similar to MTL amnesia and is part of the broader Papez circuit.

Temporal lobe epilepsy (TLE)

The most common focal epilepsy in adults. Features:

- **Mesial temporal sclerosis** is the most common pathology; hippocampal atrophy and gliosis
- **Auras** include rising epigastric sensation, déjà vu, jamais vu, fear, olfactory or gustatory hallucinations, and complex visual or auditory experiences
- **Complex partial seizures** with impaired awareness, oroalimentary or manual automatisms, and post-ictal confusion
- **Interictal memory dysfunction** is common, lateralized by side of seizure focus

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- **Geschwind syndrome** — described interictal personality features (hyperreligiosity, hypergraphia, hyposexuality, viscosity); controversial as a syndrome

Visual recognition disorders

- **Prosopagnosia** — impaired face recognition with preserved other visual recognition; bilateral or right fusiform lesions
- **Visual object agnosia** — apperceptive (cannot integrate visual features) vs. associative (cannot link percept to meaning); occipitotemporal lesions
- **Pure alexia (alexia without agraphia)** — impaired reading with preserved writing; left occipitotemporal lesion damaging the visual word form area or its connections
- **Topographic disorientation / landmark agnosia** — parahippocampal place area damage

Klüver-Bucy syndrome

Following **bilateral anterior temporal lobe damage** (including amygdala). Features:

- Placidity and reduced fear response
- Hyperorality (tendency to examine objects orally)
- Hypersexuality
- Visual agnosia ("psychic blindness")
- Hypermetamorphosis (compulsive attention to environmental stimuli)
- Dietary changes

Full syndrome is rare in humans; partial features occur after bilateral temporal damage from herpes encephalitis, TBI, or bvFTD.

Stroke syndromes

- **MCA inferior division infarct (dominant)** — Wernicke's aphasia; contralateral superior quadrantanopia (Meyer's loop involvement) may occur with deeper extension
- **MCA inferior division infarct (non-dominant)** — visuospatial deficits, prosody comprehension deficits, sometimes confusional state
- **PCA infarct (medial temporal)** — amnesia (if bilateral or dominant hippocampus), homonymous hemianopia, alexia without agraphia (left occipitotemporal)
- **Anterior choroidal artery infarct** — small lesions affecting uncus, anterior hippocampus, and amygdala can produce memory and emotional changes

Neurodegenerative localization

- **Alzheimer disease** — earliest pathology involves transentorhinal and entorhinal cortex.

hippocampal atrophy emerging early; clinically manifests as progressive amnesic syndrome

- **Semantic variant PPA (svPPA)** — bilateral anterior temporal atrophy, left-predominant; progressive semantic loss
- **Logopenic variant PPA (lvPPA)** — left posterior temporal and inferior parietal atrophy; word retrieval and sentence repetition deficits; often Alzheimer pathology
- **Behavioral variant FTD with temporal predominance (right temporal variant)** — right anterior temporal atrophy; prosopagnosia, loss of person knowledge, behavioral and emotional changes
- **Limbic-predominant age-related TDP-43 encephalopathy (LATE)** — amnesic dementia of older adults with hippocampal involvement, often mimicking AD

Other localized presentations

- **Herpes simplex encephalitis (HSE)** — predilection for medial and inferior temporal lobes and orbitofrontal cortex; produces severe amnesia and personality change
- **Limbic encephalitis** — autoimmune or paraneoplastic; bilateral medial temporal involvement with subacute amnesia, seizures, and psychiatric symptoms
- **Auditory hallucinations** — superior temporal cortex involvement in primary psychotic disorders and some seizure phenomena
- **Musicogenic epilepsy** — rare seizure type triggered by music, often with right temporal focus

Traumatic brain injury

The **anterior temporal poles** are vulnerable to contusion in closed-head injury alongside the orbitofrontal cortex, due to bony irregularities of the middle cranial fossa. Memory dysfunction, particularly for new learning, is among the most common persistent cognitive sequelae of moderate-to-severe TBI.

Assessment

Verbal memory (left MTL-sensitive)

Measure	Description
California Verbal Learning Test-3 (CVLT-3)	16-word list learning with semantic structure; learning, interference, delay, recognition

Hopkins Verbal Learning Test-Revised (HVLT-R)	12-word list, three categories; briefer alternative
Rey Auditory Verbal Learning Test (RAVLT)	15-word list learning
WMS-IV / WMS-V Logical Memory	Story recall, immediate and delayed
WMS-IV / WMS-V Verbal Paired Associates	Word-pair associative learning
NAB List Learning	Word list with three trials
RBANS List Learning / Story Memory	Brief screening
Selective Reminding Test	Verbal list learning with selective reminding methodology

Visuospatial memory (right MTL-sensitive)

Measure	Description
Brief Visuospatial Memory Test-Revised (BVMT-R)	Six geometric figures; learning and delayed recall
Rey-Osterrieth Complex Figure (memory trials)	Copy followed by immediate and delayed recall
WMS-IV / WMS-V Visual Reproduction or Designs	Geometric design recall
Continuous Visual Memory Test	Recognition-based visual memory
Doors and People Test	Verbal and visual recall and recognition
Camden Memory Tests	Recognition memory for faces, scenes, words

Recognition memory and item familiarity

- **Warrington Recognition Memory Test** — words and faces
- **Camden Memory Tests** — including Topographical Recognition Memory
- **CVLT-3 / HVLT-R / BVMT-R recognition trials** — embedded recognition measures contrast encoding with retrieval failure

Language (temporal-relevant)

- **Boston Naming Test (BNT)** — 60-item confrontation naming

- **Multilingual Naming Test (MINT)** — alternative naming measure with cultural breadth
- **Token Test** — auditory-verbal comprehension under varying syntactic demand
- **Peabody Picture Vocabulary Test-5 (PPVT-5)** — receptive vocabulary
- **Boston Diagnostic Aphasia Examination (BDAE)** — comprehensive aphasia battery
- **Western Aphasia Battery-Revised (WAB-R)** — aphasia classification and severity
- **Semantic (category) fluency** — animals, supermarket items; sensitive to temporal-semantic dysfunction (in contrast to phonemic fluency, which is more frontal)

Semantic processing

- **Pyramids and Palm Trees Test** — semantic association across pictures and words
- **Camel and Cactus Test** — semantic association
- **Word-picture matching tasks** — embedded in PPA assessment protocols
- **Famous faces and famous names tests** — person-specific semantic knowledge, sensitive to anterior temporal dysfunction

Visual recognition and face processing

- **Benton Facial Recognition Test (BFRT)** — perceptual face matching
- **Cambridge Face Memory Test (CFMT)** — face memory
- **Visual Object and Space Perception Battery (VOSP)** — separate object and space perception subtests
- **Famous Faces Test** — recognition and naming of familiar faces

Auditory processing

- **Speech-Sounds Perception Test (Halstead-Reitan)** — phoneme discrimination
- **Seashore Rhythm Test (Halstead-Reitan)** — auditory pattern perception, right temporal sensitivity
- **Dichotic listening paradigms** — assess hemispheric specialization for auditory verbal material

Social cognition (temporal-relevant)

- **Reading the Mind in the Eyes Test** — mental state inference from eye region
- **The Awareness of Social Inference Test (TASIT)** — comprehension of social cues, sarcasm, emotion
- **Emotion recognition tasks** (Ekman 60 Faces, NimStim) — facial emotion identification, amygdala-sensitive

Bedside screening for temporal dysfunction

- **Three- and five-word delayed recall** — basic episodic memory screen
- **Story recall** at bedside (e.g., the Babcock Story)
- **Naming to confrontation** — common and low-frequency object naming
- **Single-word and sentence comprehension** — point-to commands of varying complexity
- **Repetition** — single words, phrases, low-probability sentences
- **Category fluency** (animals in 60 seconds) — sensitive to semantic and MTL dysfunction
- **Famous face recognition** — quick semantic memory screen for anterior temporal function